

SpaceLoop System & Design Architecture

Comprehensive Technical Reference: Feature Subsystems, UI Token System & Color Palette

Platform Specification	Release Stack	Security Compliance	Document ID
SpaceLoop P2P Network	React 18 / Tailwind / Flask	DPDP Act 2023 / Sec 52 Easements	SL-SYS-COLOR-2026-V1

1. Executive Summary & Core Platform Flow

SpaceLoop is India's premier peer-to-peer marketplace that transforms underutilized physical spaces (residential rooms, quiet studio corners, idle offices, cafe basements, and creative workshops) into verified, hourly-bookable micro-environments. Built around a zero-hardware physical access model and instant algorithmic micro-leases, SpaceLoop connects verified seekers (students, remote workers, creators) with verified hosts.

The end-to-end user lifecycle follows a deterministic, tamper-proof state machine:

CORE FLOW: Host Premise Verification → Listing AI Pricing → Natural Language Search → Instant Micro-Lease & █100 Escrow → Dual-Handshake Check-In → Live In-Room Console → Electrical Off Check-Out & Instant Refund

2. Complete Color Theme & Design Token Architecture

SpaceLoop utilizes a high-contrast, atmospheric **dark obsidian theme** engineered for maximum visual focus, distraction-free productivity, and clear role separation between Seekers and Hosts.

Color Role	Hex Code	Tailwind Token	Semantic Usage & Subsystem
Root Canvas	#020617	bg-slate-950	Deep obsidian backdrop for entire application body & viewport.
Card Surface	#0F172A	bg-slate-900	Elevated space cards, modal dialogs, search containers.
Glass Surface	rgba(15,23,42,0.75)	.glass-panel	Sticky header navbar, floating HUD overlays with 16px blur.
Nested Controls	#1E293B	bg-slate-800	Input boxes, category pills, slider tracks, divider panels.
Subtle Border	rgba(255,255,255,0.08)	border-slate-800	Boundary outlines, table borders, card dividing lines.
Seeker Primary CTA	#4F46E5	bg-indigo-600	Primary buttons ('Find a Space', 'Book Space', 'Student SSO').
Seeker Accent / Glow	#6366F1	brand-500	AI Matchmaker badges, pulsing indicators, active filters.
Seeker Pill Highlight	#818CF8	text-indigo-400	'█ Seeker Portal' badge, navigation icons, link highlights.
Seeker Brand Glow	#4F46E5 → #8B5CF6	from-indigo-600 via-violet-600	Brand logo gradient, primary search submit button.
Host Primary Brand	#F59E0B	accent-amber	Host dashboard revenue metrics, '+ Add Space' button.
Host Highlight	#FBBF24	text-amber-400	'█ Host Portal' badge, rating stars, earnings numbers.
Host Brand Gradient	#D97706 → #EA580C	from-amber-600 to-orange-600	Host infinity logo, 'Register Property' action pill.
Success / Verified	#10B981	text-emerald-400	100% Host Verified, Aadhaar tokenized, Session Active.
Active Session Gradient	#059669 → #0D9488	from-emerald-600 to-teal-600	Mobile HUD active room pass, physical lock handshake button.
Warning / Pending	#F59E0B	text-amber-400	KYC pending review, limited AI connectivity mode.
Danger / Overstay	#F43F5E	text-rose-400	1.5x overstay penalty, electrical check-out failure, logout.

AI Concierge	#7C3AED	bg-violet-600	Floating 'Ask LoopBot' interactive assistant badge & bubble.
Primary Typography	#FFFFFF / #F8FAFC	text-white / text-slate-50	Main headlines, space card titles, hourly price figures.
Secondary Body Text	#CBD5E1	text-slate-300	Card descriptions, addresses, legal terms, host notes.
Muted Sub-labels	#94A3B8	text-slate-400	Timestamps, distance in meters, capacity specs, icons.
Subtle Captions	#64748B	text-slate-500	Input placeholders, footer disclaimers, disabled controls.

PHYSICAL ELEVATION & FLOATING SYSTEM (index.css):

SpaceLoop avoids harsh drop-shadows by utilizing a diffused multi-layer elevation system: **Base Containers** use box-shadow: 0 4px 20px -2px rgba(0,0,0,0.38); **Interactive SpaceCards** smoothly elevate by -3px on hover with box-shadow: 0 12px 28px -4px rgba(0,0,0,0.52) under a fluid 250ms cubic-bezier(0.16, 1, 0.3, 1) transition curve.

3. End-to-End Feature Architecture & Subsystems

Every feature in SpaceLoop is fully realized across the React frontend, RESTful APIs, SQLite WAL persistence, and AI integration layers:

3.1 Natural-Language AI Intent Matchmaker

Seekers can enter natural-language prompts such as 'Quiet studio for 3 near Hauz Khas under █60/hr with good WiFi'. The AI engine extracts latent intent (budget constraints, amenities, capacity, and coordinates) and runs a composite ranking algorithm combining Haversine spatial proximity (within chosen radius), tag overlap, and budget fit. Returns transparent, explainable 'Why this matches' percentage scores.

3.2 Multimodal AI Space Inspector & Pricing Engine

When a host uploads space photos or description, the AI inspector analyzes room readiness, lighting, workstation quality, and power socket density. It calculates a suitability score (0–100%) and suggests competitive hourly pricing (█35–█75/hr) grounded in neighborhood demand heuristics.

3.3 Student SSO & Academic Subsidies

Verified students and university researchers (from institutions like IIT Delhi, Pune University, DU) authenticate via institutional domain emails (.ac.in / .edu.in) to unlock automatic 15–25% subsidies on study pods and maker workshops, opening access to high-end infrastructure.

3.4 Host Premise Custody Proof (Discom + UPI Penny Drop)

To eliminate ghost listings and fake brokers, hosts must prove physical custody before listing: (1) Submission of an active state electricity distribution company (BESCOM, BSES, MSEDCL) Consumer Account (CA) number; and (2) Automated █1 UPI Penny Drop bank match verifying that the payout account name matches the electricity bill registration.

3.5 Instant Booking & AI Micro-Lease Synthesis

SpaceLoop eliminates cumbersome multi-month rental agreements. Upon reservation confirmation, the platform dynamically compiles a plain-language **Temporary Space Use License Agreement** under Section 52 of the Indian Easements Act, 1882, stipulating exact hours, occupancy limits, and permitted activities.

3.6 Dual-Handshake Physical Access (Geofence + Dynamic QR / PIN)

Zero-hardware check-in requires a two-step cryptographic handshake: (1) The seeker's smartphone GPS must verify location within a 50-meter Haversine radius of the space; and (2) The seeker must scan the host's dynamic room QR code or enter the rotating 4-digit caretaker PIN. Remote check-ins are strictly blocked.

3.7 Live In-Room Console (Digital Access Pass)

Once checked in, the seeker's screen activates a live HUD console showing session countdown timer, WiFi network credentials, real-time decibel monitor, one-tap host messaging, and instant extension requests without re-booking.

3.8 Automated █100 UPI Micro-Escrow & Computer Vision Check-Out

Every booking places a █100 security hold in platform escrow. Upon check-out, the seeker uploads an exit photograph. Our Computer Vision inspection verifies: (1) Furniture condition match $\geq 90\%$; and (2) Mandatory electrical appliance shut-off (fans, ACs, lights). If cleared, the █100 is instantly released back to the seeker's UPI account.

3.9 Objective Telemetry Index (OTI) Reputation Engine

SpaceLoop replaces subjective star reviews with mathematical telemetry: $OTI = (35\% \text{ Punctuality}) + (35\% \text{ Room Condition}) + (20\% \text{ KYC Compliance}) + (10\% \text{ Zero Disputes})$. Builds an ungameable, objective trust reputation for both guests and hosts.

3.10 Host Earnings & Monetization Calculator

An interactive financial simulator that models monthly and annual net earnings based on space category, square footage, expected hours per day, days per month, and platform fee deductions.

3.11 LoopBot 24/7 AI Platform Concierge

An always-available AI concierge floating at the bottom-right corner of the application. LoopBot assists seekers with booking advice, suggests nearby hubs, explains escrow protocols, and guides hosts through Discom verification.

3.12 Printable Zero-Hardware QR Door Pass

Hosts can generate and print high-resolution physical door passes with integrated room QR tokens and emergency caretaker contacts, enabling instant self-serve check-in without requiring smart locks or IoT hardware.

4. Statutory & Regulatory Compliance Framework

- SpaceLoop is engineered from the ground up for strict legal enforceability and privacy protection under Indian law:
- **Section 52, Indian Easements Act, 1882:** All bookings are structured strictly as a revocable personal license to occupy premises during specified hours. Tenancy, leasehold rights, and adverse possession claims under the Transfer of Property Act, 1882 are completely excluded.
 - **Digital Personal Data Protection Act, 2023 (DPDP Act):** SpaceLoop strictly adheres to DPDP Act Section 8. Raw 12-digit Aadhaar numbers are never stored in the database. Credentials are tokenized via DigiLocker and stored as salted SHA-256 hashes with masked UI display (XXXX-XXXX-4821).
 - **Overstay & Penalty Protocol:** Overstaying past the booked end-time without approved extension triggers an automatic 150% prorated surcharge and suspension of digital pass access to prevent unauthorized squatting.

5. Technical Architecture & Component Stack

TIER / LAYER	TECHNOLOGY STACK	ARCHITECTURAL ROLE
Frontend SPA	React 18 + TypeScript + Vite	Single Page Application serving responsive desktop & mobile views.
Styling Engine	Tailwind CSS + PostCSS	Utility-first design token system with custom brand/accent palettes.
Backend API	Python 3.11 / 3.14 + Flask	Modular REST blueprints (/api/v1/auth, spaces, bookings, system).
Database & Concurrency	SQLite (WAL Mode + 5s Timeout)	Zero-lock concurrency, WAL write journal, dialect-agnostic sync.
AI Routing	Groq LLaMA 3.3 → Gemini 1.5 → Rules	Multi-tier intent parsing with deterministic heuristic fallbacks.
Production Deployment	Render (Gunicorn 2 workers x 4 threads)	Edge proxy-normalized WSGI serving pre-built SPA bundles.

AUDIT & QUALITY ASSURANCE CERTIFICATION:

SpaceLoop has been verified across all 14 end-to-end automated audit subsystems (Homepage, Spaces API, OTI Index, AI Inspector, NL Matchmaker, Listing Creation, Earnings Calculator, LoopBot Concierge, Student KYC, Host Discom Check, Instant Micro-Lease Booking, Dual-Handshake Check-In, Computer Vision Escrow Release, and All Template Views). Result: **100% OPERATIONAL WITH ZERO REGRESSION FAILS.**